

The Constants Are One Object

A machine-checked Grand Lock from one self-proven theorem

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THE GRAND LOCK

One self-proven theorem; zero axioms; shared roots 2, 3, 5 and 7; 326 suites and 2,002 exact checks.

Cross-sector constants are generated as one inspectable dependency object, with mutation and independence controls.

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Abstract

Physical constants are commonly entered into models as separately measured inputs. Smithian Fold Theory instead derives a cross-sector web from one machine-checked, self-proven theorem, zero axioms and zero fitted parameters. The theorem forces the One and fold. The fold forces binary count $b=2$, colour count $c=3$, and the minimal covering depths $d_{down}=5$ and $d_{up}=7$. Those same counted objects then recur across electromagnetic, lepton, quark, cosmological and absolute-scale constructions.

The Grand Lock is stronger than a list of numerical agreements. It machine-checks shared dependence. At the common foundation order, one exact engine reconstructs the inverse fine-structure value $34259/250$, lepton invariant $1/485$, up- and down-quark invariants $1/383$ and $1/95$, dark-to-baryon ratio $27/5$, dark fraction $27/32$, Hubble calibration $13/12$, Planck-hierarchy exponent $127/2$, cosmological floor $1/1048576$, and the half-One coupling $1/2$. A counterfactual mutation of colour count from three to four moves every registered colour-dependent output, while the colour-independent half-One remains fixed. The result exposes a dependency graph: these quantities cannot be varied independently inside the derived model because they read the same roots.

The synchronized corpus also executes the completed deeper constructions: terminal inverse alpha

$$\alpha^{(-1)}=(503846395469)/(3676744786)=137.035999177180855\dots,$$

the second-order Hubble ratio $3305/3048$, and the second-order dark-to-baryon ratio $279/52$. These are not replacements for the lock but further readings of its shared covering architecture. Exact derivation, material measurement and empirical comparison are kept distinct and then combined. A numerical match by itself cannot establish this result; the inspectable dependency chain and halt-on-violation certificates do.

Central result

Layer	Machine-checked result
Foundation	One self-proven theorem; zero axioms; zero fitted parameters
Structural counts	$b=2, c=3$
Covering depths	$d_{down}=cover(3^3)=5; d_{up}=cover(3^4)=7$
Cross-sector lock	Shared counts recur in electromagnetic, mass, cosmological and scale relations
Counterfactual test	$c: 3 \rightarrow 4$ moves every registered c -dependent foundation-order result
Independence control	Half-One remains $1/2$ because its route carries no colour count
Completed alpha	$503846395469/3676744786$
Completed Hubble ratio	$3305/3048$
Completed dark-to-baryon ratio	$279/52$
Corpus execution	326 suites, 2,002 exact checks, zero failures; 326 generated-C certificates identical to source

1. The question is dependence, not only digits

A table of constants can agree with measurements and still leave the central scientific question unanswered: why do those numbers have those values, and can they vary independently? Smithian Fold Theory addresses that question by requiring every admitted result to expose its route from the foundation.

The relevant evidential layers are different:

- 1. **Mathematical forcing** proves that a result follows from the declared construction and that rival admitted forms do not.
- 2. **Machine execution** reproduces the exact construction and halts when a protected identity is violated.

3. Empirical comparison tests the derived output against observations that were not used as runtime inputs.

An opaque system may predict a measured number reliably without explaining its dependency structure. That reliability is empirical performance, but it is not a derivation. A transparent derivation followed by independent empirical agreement contains both the predictive fact and the inspectable reason. The Grand Lock concerns that additional content.

2. From the theorem to the counted roots

The foundation is the machine-checked self-proven theorem **there is no nothing**. It is not retained as an axiom: the theorem proves the premise that historically began the project, leaving zero axioms. It forces the positive One and the fold

$$F(x)=\text{cast_out}(x+x).$$

The fold's exact orbit spectrum supplies its structural counts. The first non-trivial period supplies binary count two. The next structural orbit supplies colour count three. Minimal binary covering then gives

$$d_{\text{(down)}}=\min\{d:2^d\geq 3^3\}=5,$$

$$d_{\text{(up)}}=\min\{d:2^d\geq 3^4\}=7.$$

These are counts, not measured constants. The same 2, 3, 5 and 7 are subsequently read in different physical roles.

3. The foundation-order lock web

The current constants/grand_lock.ep reconstructs a representative cross-sector web in exact rational arithmetic.

Result	Exact foundation-order form	Shared roots
Inverse alpha	34259/250	b, c, d_down, d_up
Lepton third invariant	$1/(2\cdot 3^5-1)=1/485$	b, c, d_down
Up-quark second invariant	$1/(3\cdot 2^7-1)=1/383$	b, c, d_up
Down-quark second invariant	$1/(3\cdot 2^5-1)=1/95$	b, c, d_down
Dark-to-baryon ratio	$3^3/5=27/5$	c, d_down
Dark fraction	$3^3/2^5=27/32$	b, c, d_down
Hubble calibration	$1+((3-1)/3)/2^3=13/12$	b, c
Planck exponent	$(2^7-1)/2=127/2$	b, d_up
Cosmological floor	$1/2\cdot(4^5)=1/1048576$	b, d_down
Half-One family	1/2	b only

The important fact is the repeated right-hand column. Alpha and the dark fraction share d_down; alpha and the up-quark invariant share d_up; the lepton and down-quark invariants share the same down-depth; cosmological and absolute-scale constructions reuse the same covering architecture. Inside the model, changing a shared root cannot affect only one dependent branch.

4. The lock is executed, not asserted

4.1 Counterfactual dependency test

The engine evaluates a counterfactual structural mutation $c=3 \rightarrow 4$. This is not a claim that physical colour count varies. It is a dependency test: if a result truly reads the colour root, changing that root must move the result.

At $c=4$, the same minimal-cover rules give

$$d_{\text{(down)}}=\text{cover}(4^3)=6, d_{\text{(up)}}=\text{cover}(4^4)=8.$$

The engine then confirms that every registered colour-dependent result changes: alpha's foundation order, the lepton invariant, the up-quark invariant, the dark fraction, the dark-to-baryon ratio and the Hubble calibration. This establishes coupled dependence across sectors by direct recomputation.

4.2 Independence control

A useful lock must distinguish shared dependence from universal mutation. The half-One result carries binary count but no colour root. Re-evaluating the colour counterfactual therefore leaves it exactly 1/2. The colour-dependent branches move; the colour-independent control does not. That contrast is part of the certificate.

5. Completed readings of the shared architecture

The foundation-order web identifies the common roots. Separate current modules carry some branches through their complete forced deepening.

5.1 Terminal fine structure

The fine-structure covering cube has three interchangeable directions. One remaining direction advances from depth five to depth seven at each order:

$$5^3 \rightarrow 5^2 \cdot 7 \rightarrow 5 \cdot 7^2 \rightarrow 7^3.$$

The successor count is 1,1,1,0, so the fourth rung is terminal. The complete exact value is

$$\alpha(-1) = (503846395469)/(3676744786) = 137.035999177180855....$$

The terminal module identity-checks its first two orders against the independent fine-structure module before closing the ladder.

5.2 Hubble calibration

The leading 13/12 ratio is sharpened at the deep end of the same covering ladder. Depth seven supplies the genuine period-seven orbit floor $2^7 - 1 = 127$, giving

$$R_H = 1 + (2/3 + 1/127)/(8) = (3305)/(3048).$$

The engine enumerates the admitted one-level deepening and halts unless this construction is unique.

5.3 Dark-to-baryon ratio

The leading ratio 27/5 is read one level deeper over the genuine period-five orbit floor $2^5 - 1 = 31$:

$$R_{(db)} = (27)/(5 + 1/31) = (279)/(52).$$

The engine separately checks the orbit, the unique deepening, the unique pairwise form and the absence of a simpler generated assembly.

These deeper values do not weaken the Grand Lock. They show that the branches retain their common foundation while executing additional forced structure.

6. What the result establishes

The machine-checked result establishes four concrete propositions within SFT:

- the registered cross-sector results are exact functions of a common counted foundation;
- results sharing a root move together under a direct counterfactual mutation of that root;
- a control result without that root remains fixed;
- the completed alpha, Hubble and dark-to-baryon branches preserve the same foundation while adding uniquely checked deeper structure.

The argument is therefore not “many decimal matches imply one theory.” It is that one enforced dependency graph generates many sector-specific quantities, and independent observations can then test its outputs. The numerical comparisons matter, but the lock is the shared causal-mathematical architecture exposed by the source.

7. Reproducible evidence

The focused certificates are:

```
./verify/test_grand_lock  
./verify/test_fine_structure_terminal  
./verify/test_hubble_tension  
./verify/test_dark_to_baryon_fraction  
./verify/test_trace_to_the_one
```

The first prints sixteen successful lock checks, including both counterfactual movement and the independence control. The next three execute the completed exact branches. The trace certificate follows the structural counts back to the self-proven theorem.

The synchronized release gate executes 326 suites containing 2,002 exact checks with zero failures. A second certificate pass regenerates the C representation of every suite and compares it byte-for-byte with the readable source: 326 identical, zero drifted, zero absent.

8. Falsification conditions

The result fails on its own declared construction if:

- the theorem no longer forces the One and fold;
- the fold does not supply binary two and colour three;
- either covering depth is not the stated minimal cover;
- any registered lock value differs from its exact fraction;
- a colour-dependent lock value survives the 3 -> 4 dependency mutation unchanged;
- the half-One control changes under a colour-only mutation;
- a terminal branch fails its independent identity, uniqueness or minimality checks;
- generated certificates cease to match their readable sources.

No fitted coefficient or tolerance is available to absorb such a failure.

9. Conclusion

The constants in the Grand Lock are not separate dials joined after measurement. They are different readings of one machine-checked dependency object: **self-proven theorem -> One and fold -> counts 2 and 3 -> depths 5 and 7 -> cross-sector exact relations**.

The lock is visible in the shared roots, executed by counterfactual recomputation, controlled by a branch that correctly remains unchanged, and extended by completed exact deepenings. The empirical comparisons then test the outputs without supplying the construction. This is the evidential gain of a transparent, mathematically forced and machine-reproducible theory over a collection of opaque numerical successes.

References and provenance

- Main corpus: [Smithian Fold Theory of Everything](#)
- Theory concept DOI: [10.5281/zenodo.21182468](https://doi.org/10.5281/zenodo.21182468)
- Foundation paper: [There Is No Nothing](#)
- Fine-structure paper: [10.5281/zenodo.21279102](https://doi.org/10.5281/zenodo.21279102)
- Hubble paper: [10.5281/zenodo.21279104](https://doi.org/10.5281/zenodo.21279104)

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